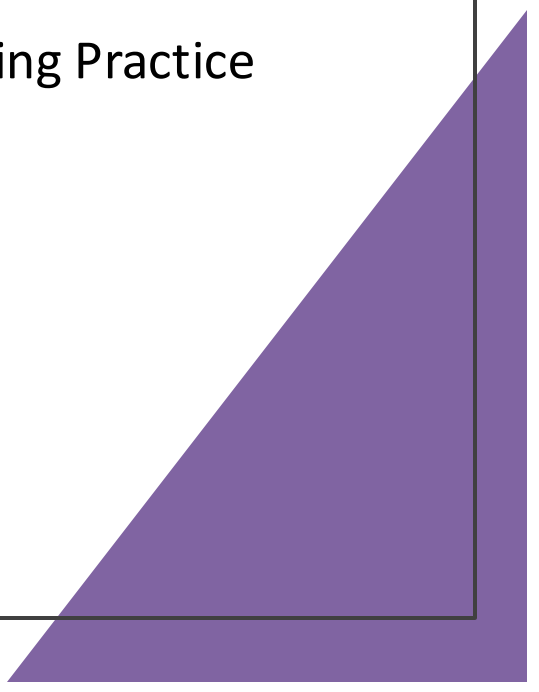


Engineering Ethics – Ethical Theories

- Understanding Moral Frameworks in Engineering Practice
 - Presented by: Dr. Razib Hayat Khan
 - Date: 15.May.2025
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Introduction to Ethics

- Ethics: A branch of philosophy dealing with morality
- Ethical engineer: Upholds moral integrity and values
- Two main categories:
 - - Consequential Ethics
 - - Non-consequential Ethics

Consequential vs. Non-Consequential Ethics

- Consequential Ethics:
 - Morality judged by outcomes (e.g., lying to save a life)
- Non-consequential Ethics:
 - Morality judged by actions themselves (e.g., lying is wrong regardless)

Consequential vs. Non-Consequential Ethics

Consequential:

Focus:

The consequences or results of an action are the primary determinant of its morality.

Example:

Utilitarianism, a common form of consequentialism, judges actions based on their ability to produce the greatest good for the greatest number of people.

Key Idea:

"The end justifies the means" - the outcome of an action determines its moral worth.

Strength:

Can justify actions that produce positive outcomes, even if they seem morally questionable in themselves.

Weakness:

Can be inflexible and may lead to outcomes that are not considered morally right.

Non-Consequential:

Focus:

The intrinsic properties of the action, such as adherence to moral rules, duties, or principles, determine its morality.

Example:

Deontology, a non-consequentialist theory, argues that certain actions are inherently wrong, regardless of their potential consequences.

Key Idea:

Moral duties and obligations must be followed, even if it leads to negative consequences.

Strength:

Provides a clear framework for moral decision-making, based on established rules and principles.

Weakness:

Can be rigid and may not always lead to the best overall outcome.

Consequential vs. Non-Consequential Ethics

Key Differences:

Focus:

Consequentialism prioritizes the outcome of an action, while non-consequentialism focuses on the action itself.

Moral Principles:

Consequentialism often leads to actions that might seem morally questionable (e.g., lying to save a life) if they produce the best overall outcome. Non-consequentialism, on the other hand, upholds principles like "never lie" or "never kill an innocent person" regardless of the consequences.

Examples:

Utilitarianism is a form of consequentialism that aims to maximize overall happiness. Deontology, a common form of non-consequentialism, emphasizes adherence to moral rules and duties.

Types of Ethical Theories

- Four key theories:
 - Golden Mean (Aristotle)
 - Rights-Based Ethics (John Locke)
 - Duty-Based Ethics (Immanuel Kant)
 - Utilitarian Ethics (John Stuart Mill)



Golden Mean – Aristotle

- Ethics as balance between excess and deficiency
- Emphasis on reason and logic
- Example: Environmental pollution – balance between industrialization and preservation

Golden Mean – Key Concepts

- Practical Wisdom: Discern the right balance
- Internal vs. External Goods:
 - Internal: experience, activity
 - External: money, status
- Over-prioritizing external goods threatens ethical integrity

Golden Mean – Challenges

- Varies with individual reasoning
- Difficult to apply consistently

Rights-Based Ethics – John Locke

- Foundation: Every person has inherent rights
- 'Live and let live' philosophy
- Protects rights to life, liberty, health, and property

Rights-Based Ethics – Key Concepts

- Human dignity is central
- Moral duty: support a just and moral society
- Laws should align with moral values

Rights-Based Ethics – Challenges

- Conflicts between individual rights (e.g., freedom vs. safety)
- Defining limits of personal rights

Duty-Based Ethics – Immanuel Kant

- Universal moral laws, no exceptions
- Moral actions are duties, not consequences
- “Act only on maxims you wish to become universal laws”

Duty-Based Ethics – Four Virtues

- Prudence: Perform duty with respect for life
- Temperance: Restrain from unethical temptations
- Fortitude: Balance between happiness and perfection
- Justice: Treat others as ends, not means

Duty-Based Ethics – Challenges

- Universal rules may not fit every situation
- Risk of rigidity and lack of flexibility

Utilitarian Ethics – John Stuart Mill

- Aim: Maximize happiness for the greatest number
- Morally right = promotes societal well-being
- Cost-benefit analysis used for decision making

Utilitarianism – Two Types

- Act Utilitarianism:
 - Evaluates each individual action
 - Problem: can justify immoral acts if outcomes are good
- Rule Utilitarianism:
 - Follows rules that generally lead to best outcomes
 - Better fit for professional settings

Utilitarianism – Engineering Context

- Example: Cost-benefit analysis for safety systems
- Professional engineers encouraged to follow Rule Utilitarianism

Engineering Theories - Differences

Golden Mean:

Focus: Finding the virtuous middle ground between two extremes of behavior.

Example: Courage lies between the extremes of cowardice and recklessness.

Key Idea: A virtuous person strives to act in accordance with the Golden Mean, avoiding both deficiency and excess.

Duty-Based Ethics:

Focus: Adhering to moral duties and following rules, regardless of consequences.

Example: The duty to tell the truth, the duty to keep promises.

Key Idea: Moral obligations are inherent and must be followed, even if they lead to undesirable outcomes.

Rights-Based Ethics:

Focus: Recognizing and protecting individual rights and freedoms.

Example: The right to free speech, the right to privacy.

Key Idea: Actions are assessed based on their respect for individual rights, regardless of the consequences.

Utilitarianism:

Focus: Maximizing overall happiness and well-being by evaluating the consequences of actions.

Example: Choosing the action that benefits the most people.

Key Idea: The most ethical action is the one that produces the greatest good for the greatest number of people.

Engineering Theories - Differences



Formulating Ethical Theories

- Ethical theories should:
 - Be logically consistent
 - Avoid contradictions
 - Be based on truth
 - Offer guidance in complex situations
 - Align with moral intuition

Uses of Ethical Theories

- Understanding moral dilemmas
- Justifying professional responsibilities
- Bridging personal and professional morality

Summary

- Ethical theories offer frameworks for moral decision-making
- Essential for engineers to uphold societal trust and responsibility
- Application requires judgment, reasoning, and empathy

Q&A / Discussion

- Invite questions
- Discuss application in real engineering scenarios

ENGINEERING ETHICS — MORAL ISSUES

Presented by: Dr. Razib Hayat Khan



WHAT IS A MORAL ISSUE?

- Involves more than technical concerns.
- Defined as: Any issue with the potential to help or harm anyone, including oneself.
- Requires ethical reasoning alongside technical decision-making.



TYPES OF MORAL ISSUES

1. Micro-ethics: Daily practice-level problems faced by engineers.

- Focused on individual conduct.

2. Macro-ethics: Broader societal or systemic issues.

- Often arise unexpectedly at national or regional levels.



EXAMPLE 1 — STRUCTURAL COLLAPSE

- Engineer learns of a marginally safe bridge.
- Superior instructs silence until next budget cycle.
- Moral Dilemma: Public safety vs. company protocol.



EXAMPLE 2 — CONFIDENTIAL DATA THEFT

- Colleague caught copying sensitive data.
- Choices: Stop friend quietly or Report to management.
- Dilemma: Loyalty vs. accountability.



EXAMPLE 3 — LOST PROTOTYPE

- Engineer loses prototype before deadline.
- Considers outsourcing to avoid job risk.
- Question: Is this morally acceptable?



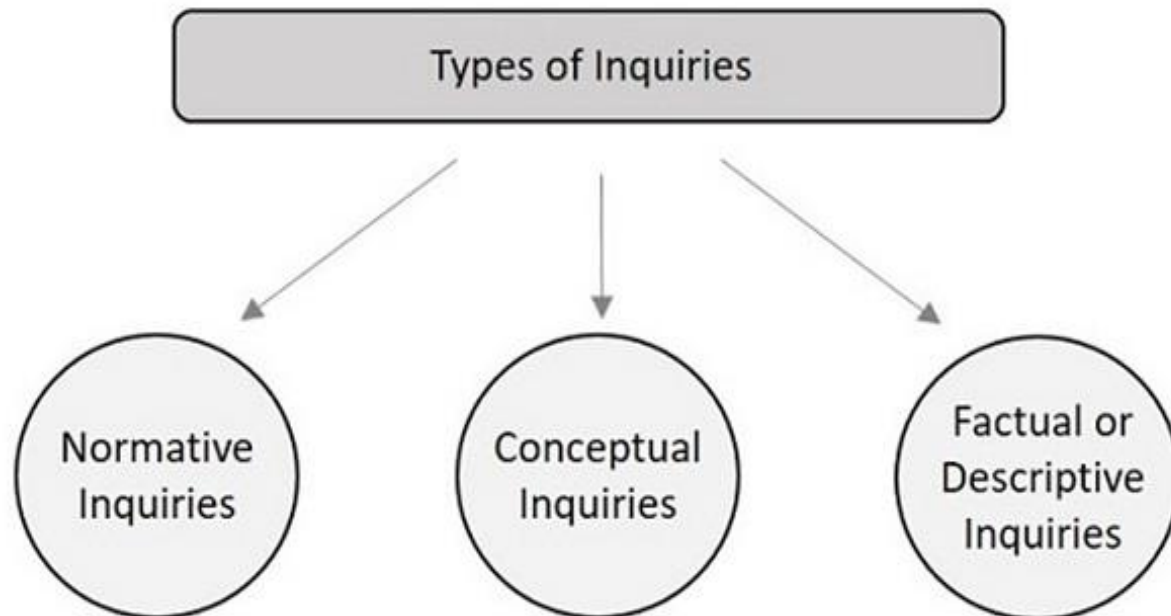
UNDERSTANDING MORAL DILEMMAS

- No single 'correct' answer.
- Solutions often require reflection and ethical judgment.
- Important to follow a slow and clear decision-making process.



TYPES OF INQUIRIES IN ENGINEERING ETHICS

- 1. Normative Inquiries
- 2. Conceptual Inquiries
- 3. Factual/Descriptive Inquiries



NORMATIVE INQUIRIES

- Concerned with 'what ought to be done.'
- Defines morally desirable behaviors and responsibilities.
- Examples: Ensuring public safety, Following legal procedures.



CONCEPTUAL INQUIRIES

- Concerned with the meaning of ethical concepts.
- Explores principles such as:
 - - Safety
 - - Professionalism
 - - Bribery and its moral implications



FACTUAL / DESCRIPTIVE INQUIRIES

- Based on scientific and empirical investigation.
- Helps in:
 - - Understanding engineering practices
 - - Assessing risks
 - - Evaluating professional effectiveness



CONCLUSION

- Engineering decisions often have ethical dimensions.
- Recognizing moral issues helps in responsible decision-making.
- Combining normative, conceptual, and factual insights leads to better ethical practice.



QUESTIONS & DISCUSSION

- Open floor for Q&A or case discussions.



ENGINEERING ETHICS – MORAL DILEMMAS

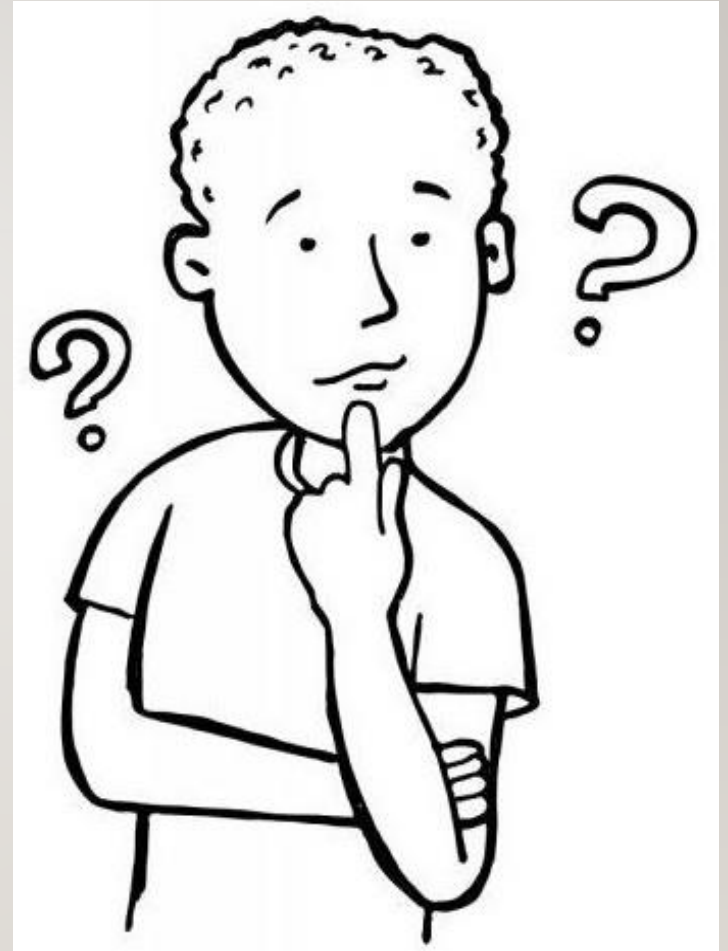
- Understanding Complex Ethical Decisions
- Presented by: Dr. Razib Hayat Khan
- Date: 16.05.2025

INTRODUCTION

- Engineering ethics involves complex decision-making.
- Moral dilemmas arise when moral reasons conflict.
- Common elements: Rights, Duties, Obligations, Goods.

WHAT IS A MORAL DILEMMA?

- A situation where making the “right” decision is difficult.
- Conflicting moral values prevent clear decisions.
- Requires critical thinking and ethical reasoning.



TYPES OF COMPLEXITIES

- Vagueness
- Conflicting reasons
- Disagreement

VAGUENESS

- Unclear if an action is right or wrong.
- Involves unwritten rules:
 - Loyalty
 - Respect
 - Confidentiality

CONFLICTING REASONS

- Multiple options with internal moral conflict.
- Priorities depend on:
 - Knowledge
 - Moral values
- Reasoning behind the choice is critical.

DISAGREEMENT

- More than one acceptable solution.
- Final choice depends on:
 - Context
 - Most probable outcome
- Requires interpretation and analysis.

STEPS TO RESOLVE MORAL DILEMMAS

1. Identification
2. Ranking
3. Inquiries
4. Discussions
5. Final Solution

STEP I – IDENTIFICATION

- Understand the issue.
- Know the duties and responsibilities.
- Identify:
 - Moral factors
 - Competing rights
 - Conflicting ideas

STEP 2 – RANKING

- List and prioritize considerations.
- Use moral aspects for ranking.
- Avoid bias or partiality.

STEP 3 – INQUIRIES

- Gather all facts.
- Explore alternative actions.
- Assess implications.

STEP 4 – DISCUSSIONS

- Involve multiple viewpoints.
- Promote dialogue and debate.
- Enable comprehensive analysis.

STEP 5 – FINAL SOLUTION

- Analyze truths and flaws.
- Consider all perspectives and facts.
- Select the most ethical resolution.

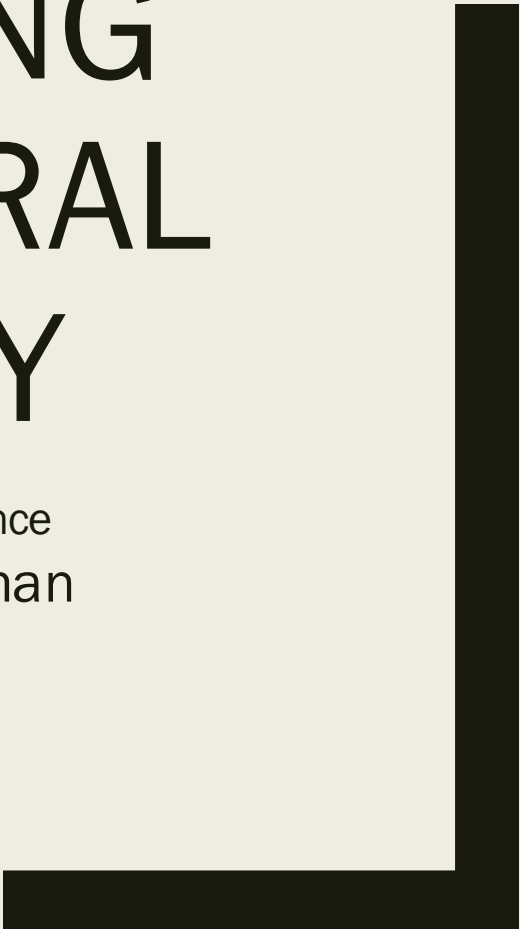
CONCLUSION

- Moral dilemmas require:
 - Structured thinking
 - Ethical frameworks
 - Collaboration
- Helps engineers make informed and ethical decisions.



ENGINEERING ETHICS: MORAL AUTONOMY

Understanding Ethical Self-Governance
Presented by: Dr. Razib Hayat Khan
Date: 16.05.2025



Introduction to Moral Autonomy

- Definition: Self-governing or self-determining actions in ethical decisions
- Key Focus: Acting independently without external distortion
- Application: Present in moral, ethical, and political philosophy

Core Concept

Moral Autonomy:

- Ability to decide right or wrong independently
- Enhances self-determination
- Guides personal conduct beyond external influences

Skills Needed for Moral Autonomy

- Relating problems to law, economics, religion
- Clarifying and understanding opposing arguments
- Suggesting solutions based on facts
- Viewing problems from multiple perspectives
- Tolerance in giving moral judgments

Skill 1 – Interdisciplinary Thinking

- Relate problems to existing laws and principles
- Differentiate moral reasons from legal or economic reasoning

Skill 2 – Analytical Clarity

- Clarify arguments against moral standards
- Judge differences and similarities in ethical concerns

Skill 3 – Fact-Based Solutions

- Suggest consistent and unbiased solutions
- Base decisions on truths and complete aspects

Skill 4 – Imaginative Perspective

- Understand issues from all viewpoints
- Develop alternative solutions with empathy

Skill 5 – Tolerant Judgement

- Provide detailed reasoning behind decisions
- Remain impartial and explain both actions done and those recommended

Improving Moral Autonomy

- Cultivate patience, interest, and humanity
- Be strict with “don’ts,” liberal with “dos”
- Practice kindness and ethical consistency

Supporting Ethical Reasoning

- Use of ethical language to support or oppose views
- Importance of tolerance, especially in conflicts
- Maintain moral honesty and understand human behavior

Conclusion

- Moral autonomy fosters ethical independence
- Essential for engineers facing complex decisions
- Continuous self-improvement is key to moral integrity